Series : 4MNKL

SET ~ 2

रोल नं.
Roll No.

प्रश्न-पत्र कोड
Q.P. Code **430/4/2**

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

नोट/ NOTE :

(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं ।
Please check that this question paper contains 23 printed pages.

(II) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(III) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 38 प्रश्न हैं ।
Please check that this question paper contains 38 questions.

(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें ।
Please write down the Serial Number of the question in the answer-book at the given place before attempting it.

(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

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गणित (बुनियादी)
MATHEMATICS (BASIC)

निर्धारित समय : 3 घण्टे
Time allowed : 3 hours

अधिकतम अंक : 80
Maximum Marks : 80

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains total 38 questions. All questions are* ***compulsory.***

(ii) *This question paper is divided into* ***five*** *Sections –* ***A, B, C, D*** *and* ***E.***

(iii) *In* ***Section–A,*** *questions number 1 to 18 are Multiple Choice Questions (MCQs) and questions number 19 & 20 are Assertion-Reason based questions of 1 mark each.*

(iv) *In* ***Section–B,*** *questions number 21 to 25 are Very Short Answer (VSA) type questions of 2 marks each.*

(v) *In* ***Section–C,*** *questions number 26 to 31 are Short Answer (SA) type questions carrying 3 marks each.*

(vi) *In* ***Section–D,*** *questions number 32 to 35 are Long Answer (LA) type questions carrying 5 marks each.*

(vii) *In* ***Section–E,*** *questions number 36 to 38 are* ***case study based*** *questions carrying 4 marks each. Internal choice is provided in 2 marks questions in each case study.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section-B, 2 questions in Section-C, 2 questions in Section-D and 3 questions in Section-E.*

(ix) *Draw neat diagrams wherever required. Take* $\pi = \frac{22}{7}$ *wherever required, if not stated.*

(x) *Use of calculator is* ***NOT*** *allowed.*

Section – A

$20 \times 1 = 20$

(Multiple Choice Questions)

Q. No. 1 to 20 are multiple choice questions of 1 mark each.

1. A card is drawn from a well shuffled deck of 52 cards. The probability of getting an ace or a ten is :

   (A) $\frac{1}{13}$ (B) $\frac{2}{13}$

   (C) $\frac{1}{26}$ (D) $\frac{4}{13}$

2. The distance between the points $(-5, 1)$ and $(2, 2)$ is :

   (A) $2\sqrt{5}$ units (B) $\sqrt{10}$ units

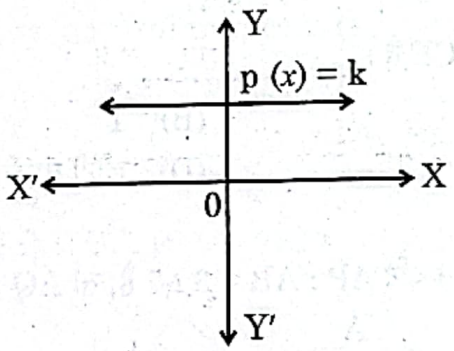
   (C) $5\sqrt{2}$ units (D) $3\sqrt{2}$ units

3. Probability of getting an irrational number at random from the numbers $\sqrt{3}, \sqrt{4}, 3\sqrt{9}, 3\sqrt{8}, \sqrt{5}, 0, 4^{\frac{3}{2}}$ is :

   (A) 0 (B) $\frac{4}{7}$

   (C) $\frac{3}{7}$ (D) 1

4. The graph of polynomial $p(x) = k$ is shown here. Number of zeroes of polynomial $p(x)$ is :



   (A) 0 (B) 1

   (C) 2 (D) infinitely many

5. Chord AB subtends an angle of 40° at the centre of the circle of radius 9 cm. The length of arc AB is :

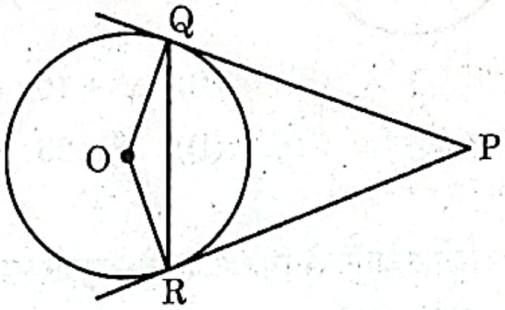
   (A) $\frac{22}{7}$ cm (B) $\frac{198}{7}$ cm

   (C) $\frac{44}{7}$ cm (D) $\frac{54}{7}$ cm

6. $7 \times 29 \times 23 + 1$ is :

(A) a prime number.
(B) divisible by 23.
(C) an odd number.
(D) a composite number.

7. In the given figure, PQ and PR are two tangents drawn to a circle with centre O. If $\angle ORQ = 25°$, then the measure of $\angle PQR$ is :



(A) 65°
(B) 25°
(C) 50°
(D) 75°

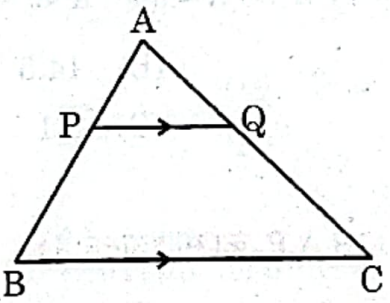
8. The value of $\tan 30° \tan 60° - \sin 90° \cos 90°$ is :

(A) 1
(B) 0
(C) 2
(D) not defined

9. HCF of two consecutive natural numbers is :

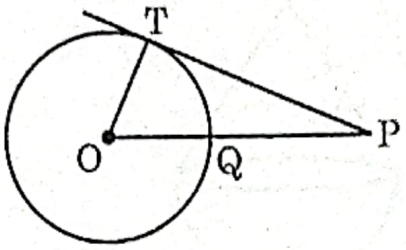
(A) 2
(B) 1
(C) 0
(D) smaller number

10. In the given figure, PQ || BC. If AP : AB = 3 : 7 then, AQ : QC equals :



(A) 3 : 7
(B) 3 : 10
(C) 7 : 3
(D) 3 : 4

11. PT is tangent to the circle with centre O and radius 5 cm. OP intersects the circle at Q. If PQ = x, then PT^2 equals :



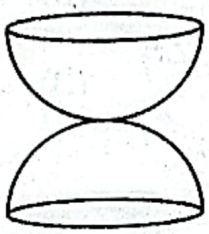
(A) $x^2 + 5x$

(B) $x^2 + 10x + 50$

(C) $x^2 + 10x$

(D) $x^2 - 25$

12. Two wooden solid hemispheres of same radii r, are joined at a point, as shown in the figure. The total surface area of the object is :



(A) $4\pi r^2$

(B) $6\pi r^2$

(C) $3\pi r^2$

(D) $5\pi r^2$

13. While calculating mean of a grouped frequency distribution using step deviation method $\left(u = \frac{x - a}{h}\right)$ it was found that $\overline{x} = 62$, $a = 47.5$, $h = 5$. The value of $\overline{u}$ is :

(A) 3

(B) 14.5

(C) 2.9

(D) 3.1

14. n^{th} term of an A.P. is $5n - 15$. The common difference of the A.P. is :

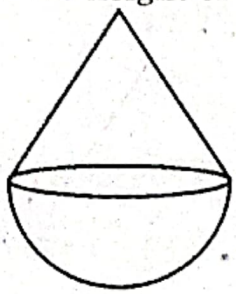
(A) 5n

(B) 5

(C) -5

(D) 10

15. A cone with slant height 10 cm and radius 6 cm is surmounted on a hemisphere of same radius. The height of the toy is :



(A) 12 cm
(B) 16 cm
(C) 14 cm
(D) 10 cm

16. n^{th} term of the A.P. $-\frac{3}{2}, \frac{3}{2}, \frac{9}{2}, \ldots$ is :

(A) $\frac{3n}{2} - 3$
(B) $3n - \frac{9}{2}$
(C) $\frac{3n-9}{2}$
(D) $3n + \frac{3}{2}$

17. If $\sec\theta = \sqrt{3}$, then the value of $\tan\theta$ is :

(A) $\sqrt{3} - 1$
(B) $60°$
(C) 2
(D) $\sqrt{2}$

18. The value of k for which the equation $5x^2 - 2x + k = 0$ has equal real roots, is :

(A) 5
(B) $-\frac{1}{5}$
(C) $\frac{1}{5}$
(D) 0

(Assertion – Reason based questions)

Directions : In questions number **19** and **20**, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option :

(A) Both, Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
(B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not correct explanation for Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** 4^n can not end with the digit zero.

Reason (R) : Prime factorisation of 4^n is unique.

20. **Assertion (A) :** The probability of an event can not be $\frac{1}{0.9}$.

Reason (R) : $0 \leq P(E) \leq 1$ for an event E.

Section – B

(Very Short Answer Type Questions) **5 × 2 = 10**

Q. Nos. **21** to **25** are Very Short Answer type questions of **2** marks each.

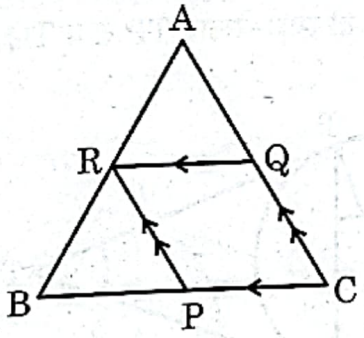
21. If $\sin 2A = \frac{\sqrt{3}}{2}$ and $2 \tan B + 1 = 3$, then find the value of $(A + B)$.

22. (a) If α, β are the zeroes of polynomial $p(x) = -9x^2 - 6x + 1$. Find the value of $\alpha^2 + \beta^2$.

OR

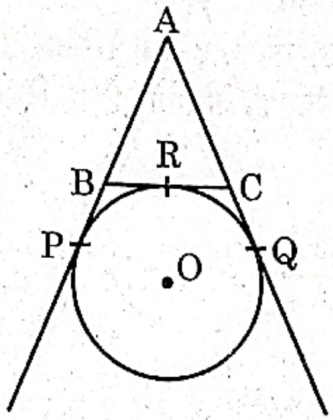
(b) Form a quadratic polynomial whose zeroes are twice the zeroes of polynomial $p(x) = x^2 - 3x - 5$.

23. In the given figure, QR || CB and RP || AC. If BR = 10 cm, QA = 12 cm, BP = 12 cm and PC = 18 cm, then find the lengths of AR and QC.



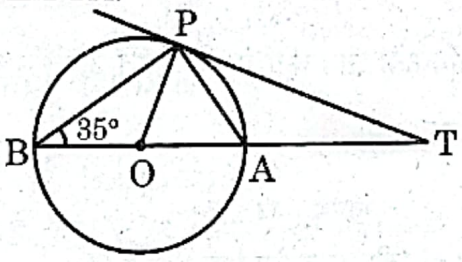
24. A (– 2, 3) and B (4, 1) are end points of the diameter of a semi-circle. The semi-circle intersects y-axis at point P. Find the co-ordinates of the point P.

25. (a) In the given figure, AP, AQ and BC are tangents to the circle with centre O. If AB = 6 cm, AC = 7 cm and BC = 5 cm, then what is the length of AP ?



OR

(b) In the given figure, TP is tangent to a circle with centre O. Diameter BA when produced meets the tangent at T. If $\angle ABP = 35°$, then find the measure of $\angle PTA$.

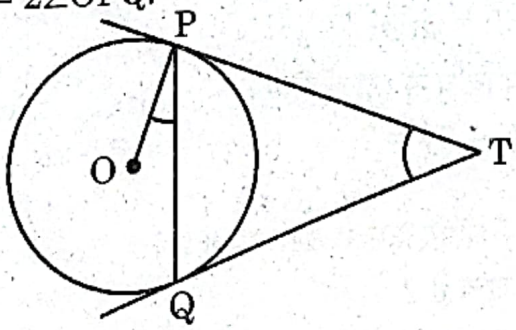


Section – C

(Short Answer Type Questions) **6 × 3 = 18**

Q. Nos. **26** to **31** are Short Answer type questions of **3** marks each.

26. Two tangents TP and TQ are drawn to a circle with centre O, from an external point T.
Prove that $\angle PTQ = 2\angle OPQ$.



27. Prove that $\sqrt{5}$ is an irrational number.

28. (a) The vertices of a rhombus ABCD are A(– 3, – 4), B(5, – 3), C(1, 4) and D(– 7, 3). Find the length of both the diagonals. Hence, find area of the rhombus ABCD.

OR

(b) The line segment joining the points A (– 5, 1) and B (7, 6) is trisected at the points P and Q such that P is nearer to A. If P lies on the line $x + y = k$, then find the value of k.

29. (a) Prove that $\frac{(\sec\theta + \tan\theta)^2 - 1}{(\sec\theta + \tan\theta)^2 + 1} = \sin\theta$.

OR

(b) Prove that $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = 1 + \sec A \operatorname{cosec} A$.

30. A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the corresponding (i) minor segment (ii) major segment.

31. Two dice are rolled together. Find the probability that at least one of the numbers obtained is a multiple of 3.

Section – D

(Long Answer Type Questions) **4 × 5 = 20**

Q. Nos. **32** to **35** are Long Answer type questions of **5** marks each.

32. (a) The difference between two numbers is 12. The greater number is 6 less than twice the smaller one.
   (i) Representing the above situation, frame two linear equations in two variables.
   (ii) Show that the equations have unique solution.
   (iii) Solve the equations and hence find the numbers.

OR

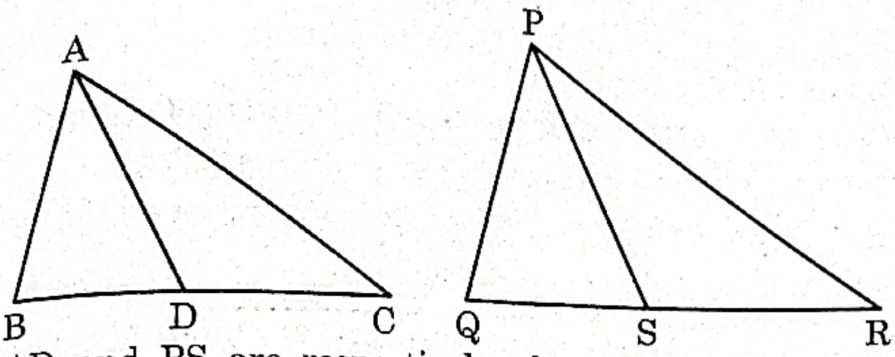
(b) Solve the following equations graphically :
   $x + y = 7$ and $2x - 5y = 7$

33. The third and ninth term of an A.P. are 4 and – 8 respectively.
   (i) Which term of the A.P. is zero ?
   (ii) Find the value of n if $S_n = -36$.

34. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

OR

(b)



AD and PS are respectively, the medians of ΔABC and ΔPQR. If $\Delta ABC \sim \Delta PQR$, then prove that

(i) $\Delta ADC \sim \Delta PSR$

(ii) $\frac{AD}{PS} = \frac{BC}{QR}$

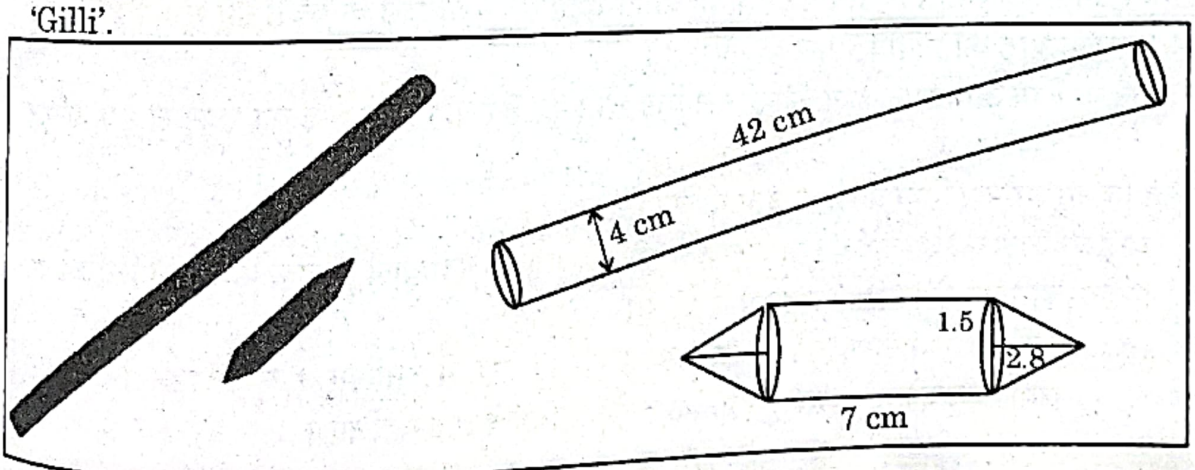
35. A statue, 2 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the bottom of the statue is 45°. Find the height of the pedestal and its distance from the point of observation on the ground. (use $\sqrt{3} = 1.73$)

Section – E

(Case-study based Questions) **3 × 4 = 12**

Q. Nos. **36** to **38** are Case-study based Questions of **4** marks each.

36. 'Gilli Danda' is a very popular traditional game of India which is played with two wooden sticks – the larger one is called 'Danda' and smaller one 'Gilli'.



'Danda' – It is cylindrical in shape with diameter 4 cm and length 42 cm.
Gilli – It is cylindrical in middle with identical conical ends of same radius 1.5 cm and length 2.8 cm. The length of cylindrical part is 7 cm.
Based on the above, answer the following questions :

(i) Find the volume of wood used in making both the conical parts of Gilli. 1

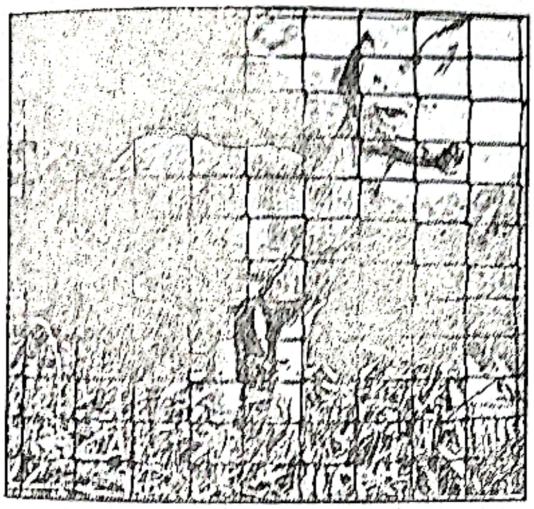
(ii) Find the volume of wood used in making cylindrical part of Gilli. 1

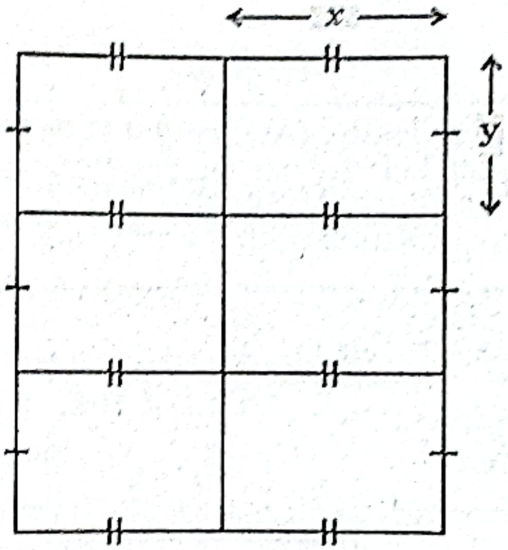
(iii) (a) A cylindrical log of wood of radius 1.5 cm and length 14 cm is used to make Gilli. Find the volume of the wood scrapped. 2

OR

(b) Find the total surface area of 'Danda'. 2

37.





Observe the figure given above. It shows six identical rectangular enclosures made by using fencing wire mesh. These enclosures are used to protect baby animals in a zoo. Dimensions of each enclosure is x feet × y feet. The total length of fencing required is 152 feet and area of each enclosure is 80 square feet.

Based on the above, answer the following questions :

(i) Write an expression for length of fencing required in terms of x and y. 1

(ii) Write the area of each enclosure in terms of x. 1

(iii) (a) Write the above equation in quadratic equation form and thus find the dimensions of each enclosure using factorisation method. 2

OR

(b) Using above equation in quadratic form, solve the equation and find the dimensions of each enclosure using quadratic formula. 2

38. **CENTRAL POLLUTION CONTROL BOARD'S AIR QUALITY STANDARDS**

AIR QUALITY INDEX (AQI)	CATEGORY
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-400	Very Poor

The Air Quality Index (AQI) is a scale from 0 to 500 that indicates air quality, with higher numbers signifying more pollution and greater health concerns.

Mansi collected the daily data of AQI of her city for a month and presented it as given below :

AQI Range :	1 – 100	101 – 200	201 – 300	301 – 400	401 – 500
Number of Days :	3	9	12	4	2

(i) Convert the data to continuous frequency distribution. 1

(ii) What is the quality of air in most of the days of the month ? 1

(iii) (a) Using table formed in part (i), find mode of the data. 2

OR

(b) Using table formed in part (i), find median of the data. 2